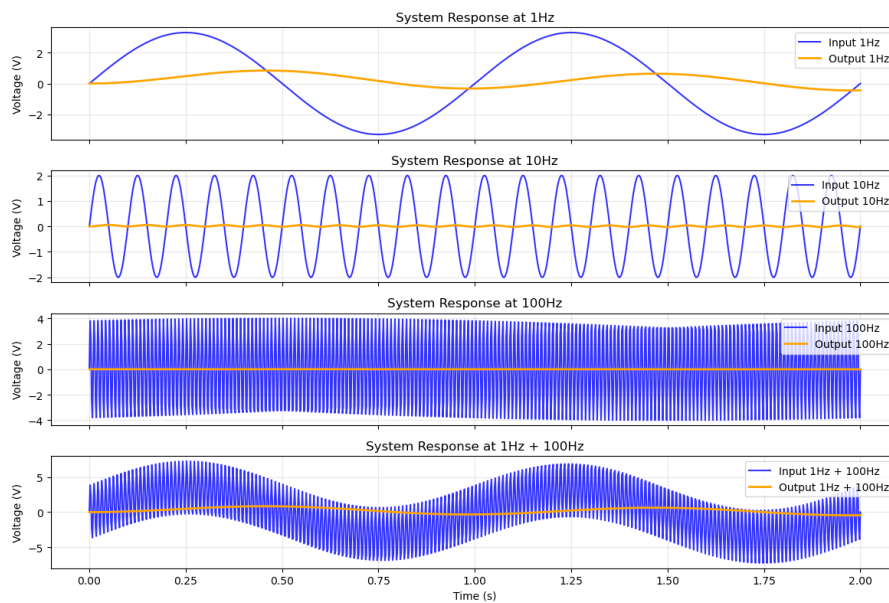
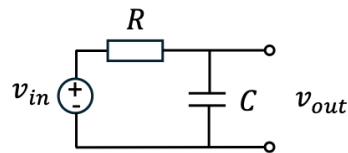


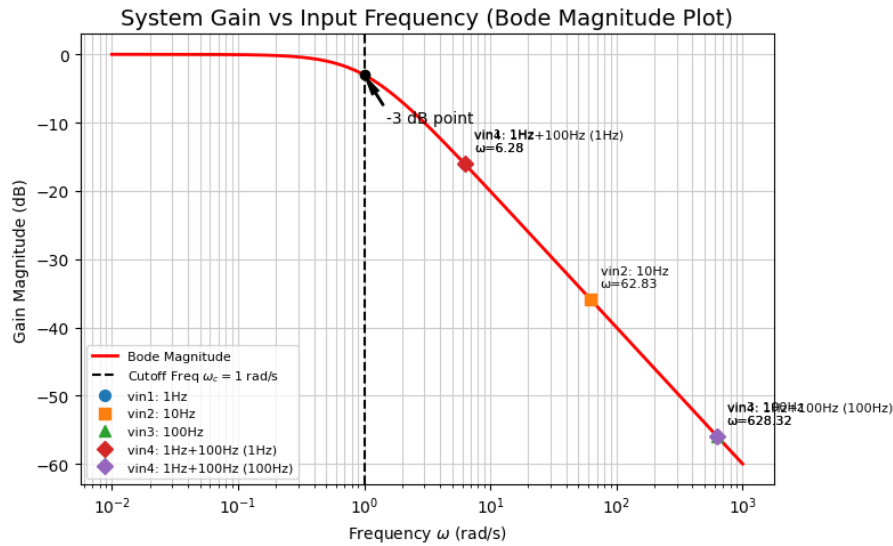
Sensor HW4

March 27, 2026

- For the following circuit with $R = 1\ \Omega$ and $C = 1\ F$. Please plot the system response v_{out} with the following four case inputs:
 - $v_{in} = 3.3 \sin(2\pi t)$;
 - $v_{in} = 2 \sin(10 * 2\pi t)$;
 - $v_{in} = 4 \sin(100 * 2\pi t)$;
 - $v_{in} = 3.3 \sin(2\pi t) + 4 \sin(100 * 2\pi t)$; Please explain the phenomenon you observed.
- Please plot the system gain as a function of the input frequency.



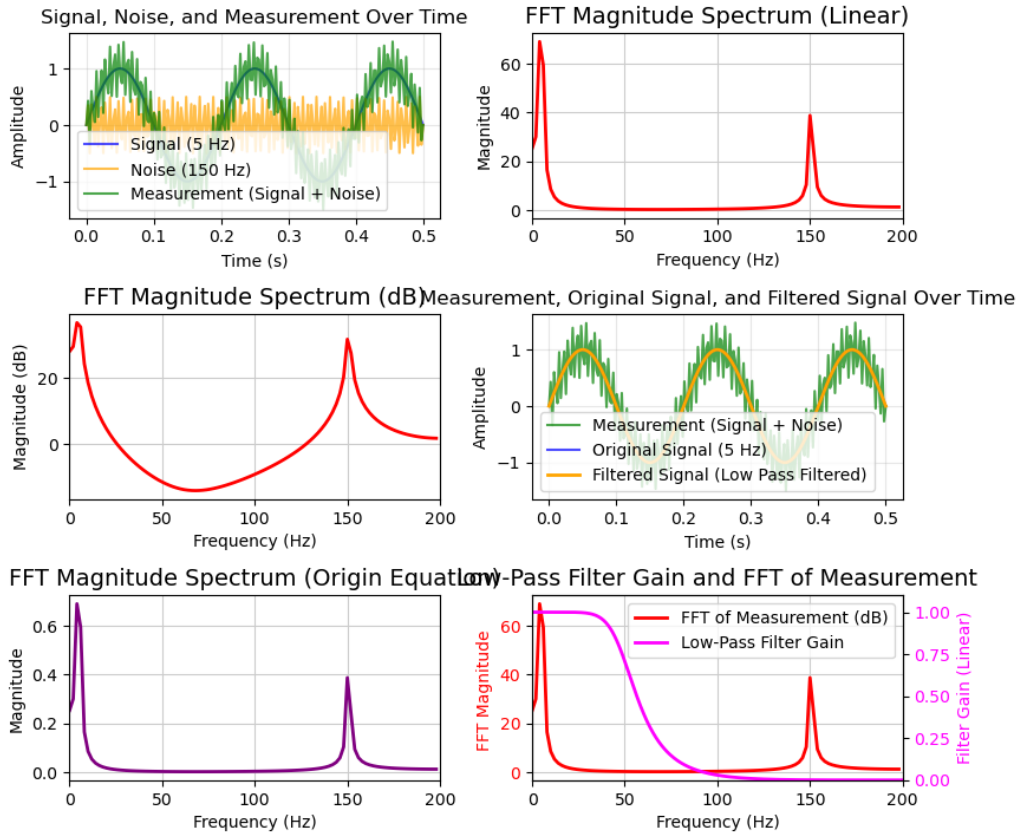
We use the `signal.bode` function to find the system gain.



2. A signal of $\sin(10\pi t)$ is interference by a noise of $0.5\sin(300\pi t)$. We plot the signal, noise, and measurement in Sub-figure 1. The FFT result in Sub-figure 2. The FFT result in dB scale in Sub-figure 3. The measurement, the signal, and the result after low pass filter in Sub-figure 4. Please also do a FFT with its original equation:

$$y[k] = \sum_{n=0}^{N-1} e^{-2\pi j \frac{kn}{N}} x[n]$$

And plot the result in Sub-figure 5. Please plot the Gain of the low-pass filter as well as the FFT of the measurement in Sub-figure 6.



3.

Table Prob. 1 gives 50 sample values of a random signal.

- (a) Estimate the mean and standard deviation of the signal.
 (b) Using an interval $\Delta y = 0.5$ V, draw the P_j and C_j discrete probability distributions for the signal.

-0.59	1.02	-0.25	-0.34	0.95	1.24	-0.30	0.21	-0.89
-1.00	1.36	0.03	0.04	-0.13	-0.71	-1.23	0.03	-1.00
0.65	0.11	0.99	0.17	0.39	2.61	-0.08	-0.33	0.99
2.15	0.91	0.89	1.43	-1.69	-0.25	2.47	-1.97	-2.26
0.42	0.05	0.26	0.33	-0.42	0.79	-0.07	-0.32	-0.66
-0.63	-0.06	-0.61	0.77	1.90				

(a)

the mean value: $\bar{y} = \frac{1}{50} \sum_{i=1}^{50} y_i = 0.1474$

the standard deviation: $s = \sqrt{\frac{1}{50-1} \sum_{i=1}^{50} (y_i - \bar{y})^2} = 1.0218$

(b)

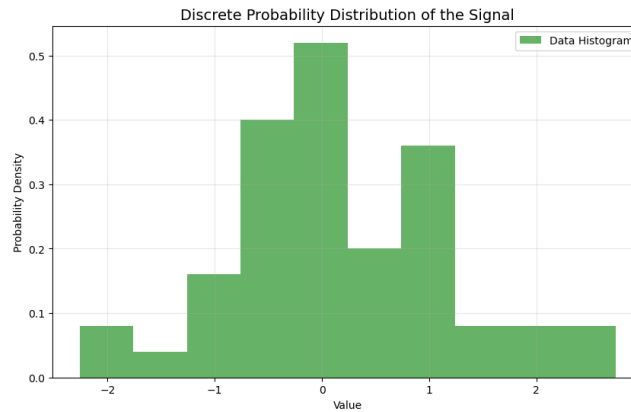


Figure 1: Probability distribution

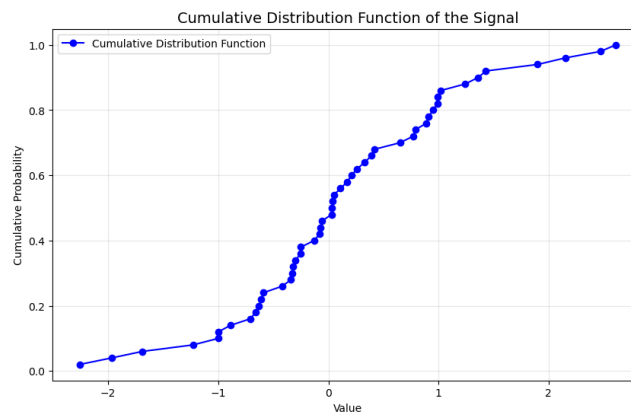


Figure 2: Cumulative distribution